TEAM-15

Ensuring Fairness & Detecting Bias in Automated Decisions

Problem Statement :

What : Automated systems in hiring, lending & judicial decisions are inheriting and amplifying human biases from historical training data — causing systemic discrimination against specific demographics (race, gender, age).

How : Build an auditable AI pipeline using Google Vertex AI + Fairlearn that detects bias pre-training and applies fairness constraints during model training — giving every model an instant Fairness Grade.

Why it is useful : Millions of life-changing decisions (loan approvals, job shortlisting, bail verdicts) are made by biased algorithms. This system embeds Code Justice into AI — making equitable decisions a technical requirement, not an afterthought.

Abstract :

An end-to-end Fairness-Aware Machine Learning framework that audits AI models for hidden bias and mitigates unfairness using pre-processing re-weighting and in-processing fairness constraints — ensuring equalized odds across all protected groups.

Short Explanation : Upload any ML model → system scans for bias → applies fixes → outputs a Fairness Grade (A/B/C/F) with full XAI explainability report showing exactly why each decision was made.

Tech Stack Used :

- Frontend UI : React.js — Real-time Bias Audit Dashboard with live fairness metrics
- Backend Logic : Python + FastAPI + Scikit-learn + Fairlearn library
- Database / Store : SQLite
- Google Technology : Google AI + Cloud DLP (bias detection) + Cloud Natural Language API (unstructured data bias testing)

Architecture :

System Workflow — Login → Register → Upload Dataset → Audit → Fix → Dashboard

① **USER ACCESS — Login / Sign Up / Registration**
User creates account → logs in → lands on Bias Audit Dashboard

② **DATA INGESTION — Google BigQuery + Cloud Storage**
User uploads dataset → stored securely in BigQuery → metadata indexed for audit

③ **BIAS AUDIT — Google Cloud DLP (Pre-Processing)**
Sensitive attribute detection → Statistical parity measurement → Fairness Score generated

④ **MITIGATION ENGINE — Google AI**
Data re-weighting → Demographic parity constraints → Equalized odds model training

⑤ **AUDIT DASHBOARD — React.js Frontend**
Fairness metrics visualization → False positive rate comparison → XAI explainability report → Fairness Grade (A/B/C/F)

⑥ **CONTACT & EXPORT**
Share Fairness Report → Notify stakeholders → Export audit log as PDF

Key Impact & Future Scope :

- Automated Bias Scoring — instant Fairness Grade for any uploaded model
- Transparency — Explainable AI (XAI) reports showing why each decision was made
- Embeds Code Justice directly into the DevOps/MLOps cycle
- Future: Browser plugin to audit any live AI system in real-time
- Future: Government adoption for judicial & public sector AI auditing